

JEE MAIN REVISION MASTER: DIFFERENTIATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 38 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Implicit, Logarithmic & Higher Order Derivatives	23 PYQs	Advanced Calculus
Very High	Standard Derivatives & Inverse Trigonometric Substitutions	15 PYQs	Differentiation Techniques

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Implicit, Logarithmic & Higher Order Derivatives	$\frac{d}{dx}[u^v] = u^v \left[\frac{v}{u} \frac{du}{dx} + \ln u \frac{dv}{dx} \right]$, $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt}$
Standard Derivatives & Inverse Trigonometric Substitutions	$\frac{d}{dx}[\sin^{-1} \frac{2x}{1+x^2}] = \frac{2}{1+x^2}$, $\frac{d}{dx}[\cos^{-1}(4x^3-3x)] = -\frac{3}{\sqrt{1-x^2}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Implicit, Logarithmic & Higher Order Derivatives (Very High Yield — 23 PYQs)

Core Principle: Implicit differentiation: differentiate both sides with respect to x . Logarithmic differentiation for $y = f(x)^{g(x)}$ or products/quotients. Second order derivatives (d^2y / dx^2). For parametric equations $x = f(t)$, $y = g(t)$, $(d^2y / dx^2) = (d / dt) \left((dy / dx) \right) (1 / dx/dt)$.

■ **Caution / JEE Pitfall:** For parametric second derivatives, remember the final chain rule step $\times (dt / dx)$!

■ **Speed Trick:** For homogeneous implicit equations $x^p y^q = (x+y)^{p+q}$, $(dy / dx) = (y / x)$, and $(d^2y / dx^2) = 0$.

Representative Source PYQs:

- [2026 Shift] Q#4 (MCQ): "If $\sqrt{(x)}\sqrt[3]{y} = (x+y)^n$ and $x(dy / dx) - y = 0$, then $n = \dots$ " [Ans: C]
- [2025 Shift] Q#1 (MCQ): "If $y = (\cos x / 1 + \sin x)$, then (a) $(dy / dx) = (-1 / 1 + \sin x)$ (b) $(dy / dx) = (1 / 1 + \sin x)$ (c) $\frac{dy}{dx} = -\frac{1}{2} \dots$ " [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "If $y = a \sin^3 t, x = a \cos^3 t$, then (dy / dx) at $t = (3\pi / 4)$ is..." [Ans: N/A]

3.2 Standard Derivatives & Inverse Trigonometric Substitutions (Very High Yield — 15 PYQs)

Core Principle: Differentiation of composite functions and inverse trigonometric forms using standard substitutions: $x = \tan \theta$ for $\sin^{-1}(2x / (1+x^2)) = 2\tan^{-1}x$; $x = \cos \theta$ for $\cos^{-1}(2x^2-1) = 2\cos^{-1}x$ and $\cos^{-1}(4x^3-3x) = 3\cos^{-1}x$.

■ **Caution / JEE Pitfall:** Check the given interval of x before writing the simplified inverse trigonometric form.

■ **Speed Trick:** Derivative of $u(x)$ with respect to $v(x)$ is simply $(du/dx) / (dv/dx)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The second order derivative of $\cos^{-1}(4x^3-3x)$ with respect to $\cos^{-1}(2x^2-1)$, where $(1/2) < x < 1$ is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "If $f(x) = \sin^{-1}(\frac{2x}{1+x^2})$, then $f'((1/2)) = \dots$ " [Ans: A]
- [2026 Shift] Q#3 (MCQ): "If $y = \sqrt{3-4x^2}$, then $(dy/dx) = \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The second order derivative of $\cos^{-1}(4x^3-3x)$ with respect to $\cos^{-1}(2x^2-1)$, where $(1/2) < x < 1$ is

- (A) 0
- (B) $-\frac{1}{\sqrt{1-x^2}}$
- (C) $(3/2)$
- (D) $-(3/2)$

Correct Answer: None

Solution Analysis: Take $y = \cos^{-1}(4x^3-3x)$ and $t = \cos^{-1}(2x^2-1)$ We need (d^2y/dt^2) Since $(1/2) < x < 1$ where $0 < \theta < (\pi/3)$ Now use the standard identities: $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ and $2\cos^2\theta - 1 = \cos 2\theta$ So, $y = \cos^{-1}(\cos 3\theta)$ Since $0 < 3\theta < \pi$, we get $y = 3\theta$ Also, $t = \cos^{-1}(\cos 2\theta)$ Since $0 < 2\theta < \pi$, we get $t = 2\theta$ Thus, $y = 3\theta$, $t = 2\theta$ So, $y = (3/2)t$ Hence, $(dy/dt) = (3/2)$ and $(d^2y/dt^2) = 0$ So the correct o

Benchmark Problem 2: KCET 2026 (Q#4)

If $\sqrt[3]{x}\sqrt[3]{3} = (x+y)^n$ and $x(dy/dx) - y = 0$, then $n =$

- (A) 1
- (B) $(6/5)$
- (C) $(5/6)$ [Correct]
- (D) $(4/9)$

Correct Answer: C

Solution Analysis: Given, $\sqrt[3]{x}\sqrt[3]{3} = (x+y)^n$ and $x(dy/dx) - y = 0$ From the second equation, $x(dy/dx) = y$ So, $(dy/dx) = (y/x)$ Now take logarithm on both sides of $\sqrt[3]{x}\sqrt[3]{3} = (x+y)^n$ We get $\log(x^{1/3}y^{1/3}) = \log((x+y)^n)$ Using laws of logarithms, $(1/3)\log x + (1/3)\log y = n\log(x+y)$ Differentiate both sides with respect to x : $(1/3x) + (1/3) \cdot (1/y)(dy/dx) = n \cdot (1+(dy/dx)/(x+y))$ Now substitute $(dy/dx) = (y/x)$ So the left side becomes $\frac{1}{3x} + \frac{1}{3} \cdot \frac{1}{y} \cdot \frac{y}{x} = \frac{1}{x}$